

Computer Vision

Activity

1) * Scenario :

• CCTV image contains salt and pepper noise due to transmission errors

* Techniques used :- Median filter

* Reason :- Removes salt and pepper noise effectively while preserving image edges.

2) * satellite image is corrupted by Gaussian noise from sensors.

A: * Scenario :- Satellite image is corrupted by Gaussian noise from sensors.

* Techniques Used :- Gaussian filter.

* Reason :- Smooths Gaussian noise by averaging nearby pixel values.

3) A medical x-ray image has random noise but fine edges must be preserved.

A: * Scenario :- Medical x-ray image has random noise but fine edges must be preserved.

* Techniques Used :- Bilateral filter.

* Reason :- Reduces noise while preserving important edges and fine details.

4) A scanned document has black and white dots scattered randomly.

* Scenario :- Scanned document has black and white dots scattered randomly.

* Techniques Used: Median filter.
* Reason: Eliminates isolated black and white dots without blurring text.

5) A low-light image shows heavy grain without clear boundaries?

A:- * Scenario: Low-light images shows heavy grain without clear boundaries.

* Techniques Used: Gaussian filter.

* Reason: Reduces grainy noise and produces a smoother image.

6) A foggy road image has very low contrast?

A:- * Scenario: Foggy road image has very low contrast.

* Technique Used: Histogram Equalization.

* Reason: Improves Contrast, making the road and objects more visible.

7) An underwater image appears dull and washed out?

A:- * Scenario: Underwater image appears dull and washed out.

* Technique Used: Contrast stretching

* Reason: Enhances brightness and contrast to improve image appearance.

8) A chest X-ray shows poor visibility of internal structures?

A:- * Scenario: Chest X-ray shows poor visibility of internal structures.

* Techniques Used: Adaptive Histogram Equalization.

* Reason: Enhances local contrast and reveals internal structures clearly.

9) A night-time traffic image lacks intensity variation?

A:- * Scenario: Night time traffic image lacks intensity variation.

* Techniques Used: Histogram Equalization.

* Reason: Increases intensity variation and improves overall visibility.

10) A gray scale images uses only a narrow intensity range?

A:- * Scenario: Grayscale images uses only a narrow intensity range.

* Techniques Used: Contrast stretching.

* Reason: Expands the intensity range, improving image contrast and clarity.

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